

Preistorie, Istorie, Prezent, Viitor

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Preistorie | Originile Internetului

- Dorinta de a comunica (si la distanta)
- Primele instrumente: limbajul si vorbirea
- Forma de comunicare: scrisa



scrierea pe pietre, peretii pesterilor, papirus 😊

- Comunicarea la distanta?
 - Semnale cu foc
 - Codul morse
- **Secolul 19:** electricitatea si telegraful

Istoric | Originile Internetului

1837 Charles Babbage proiecteaza "primul computer"



1863 Jules Verne:

"photo-telegraphy allowed any writing, signature or illustration to be sent faraway - every house was wired"

1891 – Ada Lovelace – primul program pentru masina analitica a lui Babbage

1936 Alan Turing – descrie masina Turing

1939 HP este fondat (Bill Hewlett, David Packard)

Istoric | Originile Internetului

1961-1972: primele incercari de comunicare prin schimbare de pachete

1961: Kleinrock – propune un model teoretic

1964: Baran – implementare de comunicare intre computere ale armatei americane

1967: ARPAnet este proiectat de Advanced Research Projects Agency

1969: primul nod operational ARPAnet, retea de 4 calculatoare

Istoric | Originile Internetului

1972:

- demonstratie publica a tehnologiilor ARPAnet
- NCP (Network Control Protocol) primul protocol host-host
- Primul program de posta electronica (e-mail)
- Se introduce simbolul @
- ARPAnet are 15 noduri

Istoric | Originile Internetului

1972-1980: Apare conceptul de *Internetworking*. Apar rețele aflate în proprietate privată.

1973: DARPA (Defense Advanced Research Projects Agency) – rețele interconectate; Robert Metcalf (Harvard) dezvoltă tehnologia Ethernet care permite transferul de date pe cablu coaxial

1974: Cerf și Kahn – propun protocolul de comunicare TCP (Transmission Control Protocol)

1978: suita de protocoale TCP/IP este standardizată via documentele RFC (Request For Comments)

La sfârșitul anilor '70: arhitecturi de rețea proprietare: DECnet, SNA, XNA

1979: ARPAnet are 200 de noduri

Istoric | Originile Internetului

1980-1990: noi protocoale, crește numărul de rețele, Internet

1983: este folosit TCP/IP

1982: este definit SMTP (Simple Mail Transfer Protocol)

1983: apare DNS (translatarea din numele hostului în adresa IP)

1985: apare protocolul FTP (File Transfer Protocol)

1986: coloana vertebrală a **Internetului**

1988: se introduce mecanismul de control al congestiei pentru TCP

Istoric&Prezent | Originile Internetului

**1990 ->2000->2022: World Wide Web, noi aplicatii
comerciale ale retelelor**

Inceputul anilor '90:

- World Wide Web**

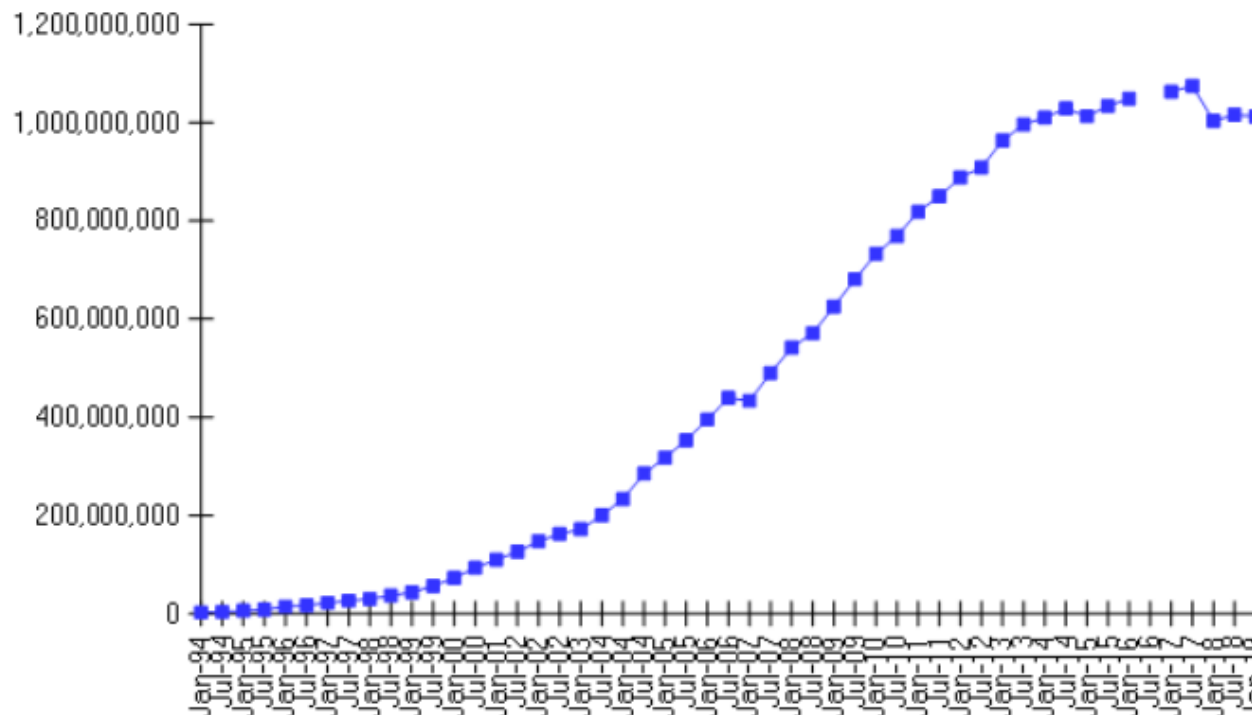
Dupa sfirsitul anilor '90 -> prezent :

- Retele mondiale
- Retele fara fir – *wireless*
- Aplicatii comerciale ale Internetului
instant messaging, P2P file sharing
retele sociale, video-streaming,
cloud-computing, fog/edge computing, 5G,
...

quantum internet prototypes (Europa, US), Micius –
satelit quantum

Prezent

Internet Domain Survey Host Count



Source: Internet Systems Consortium (www.isc.org)

Figura. Creșterea numărului de *host*-uri din Ianuarie 1994 până în Ianuarie 2019

Sursa: <https://www.isc.org/network/survey/> (IPv4)

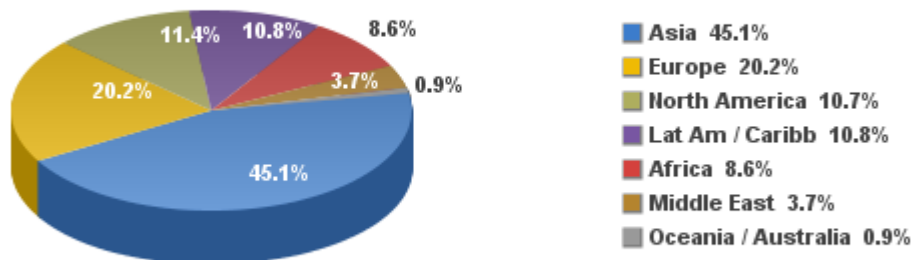
Trecut

WORLD INTERNET USAGE AND POPULATION STATISTICS December 31, 2013

World Regions	Population (2014 Est.)	Internet Users Dec. 31, 2000	Internet Users Latest Data	Penetration (% Population)	Growth 2000-2014
Africa	1,125,721,038	4,514,400	240,146,482	21.3 %	5,219.6 %
Asia	3,996,408,007	114,304,000	1,265,143,702	31.7 %	1,006.8 %
Europe	825,802,657	105,096,093	566,261,317	68.6 %	438.8 %
Middle East	231,062,860	3,284,800	103,829,614	44.9 %	3,060.9 %
North America	353,860,227	108,096,800	300,287,577	84.9 %	177.8 %
Latin America / Caribbean	612,279,181	18,068,919	302,006,016	49.3 %	1,571.4 %
Oceania / Australia	36,724,649	7,620,480	24,804,226	67.5 %	225.5 %
WORLD TOTAL	7,181,858,619	360,985,492	2,802,478,934	39.0 %	676.3 %
					806.0%

Tabel. Statistici privind utilizarea internetului în raport cu numărul de persoane aflate în principalele zone geografice ale lumii

Internet Users in the World Distribution by World Regions - 2013 Q4



Source: Internet World Stats - www.internetworldstats.com/stats.htm
Basis: 2,802,478,934 Internet users on Dec 31, 2013
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Figura. Utilizatorii de Internet în lume - împărțirea pe zone geografice
Iunie 2013 | Sursa:
<http://www.internetworldstats.com/>

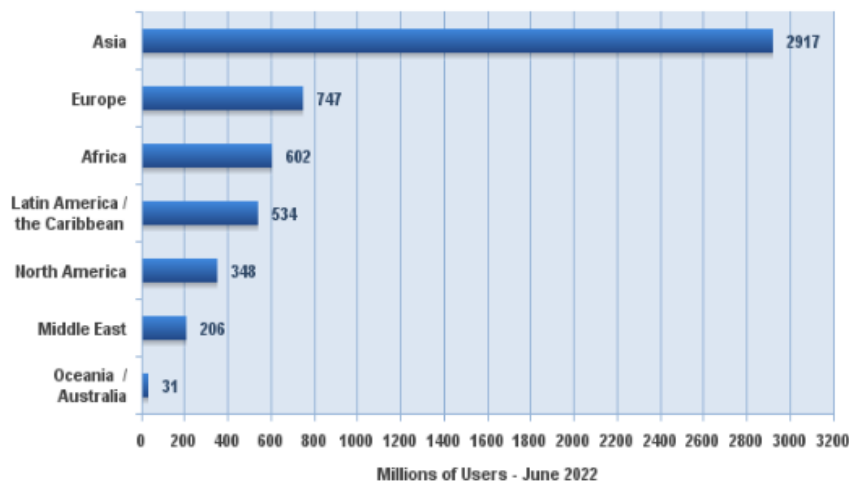
Prezent

**WORLD INTERNET USAGE AND POPULATION STATISTICS
2022 Year Estimates**

Tabel. Statistici privind utilizarea internetului în raport cu numărul de persoane aflate în principalele zone geografice ale lumii

World Regions	Population (2022 Est.)	Population % of World	Internet Users 30 June 2022	Penetration Rate (% Pop.)	Growth 2000-2022	Internet World %
Africa	1,394,588,547	17.6 %	652,865,628	46.8 %	14,362 %	11.9 %
Asia	4,352,169,960	54.9 %	2,934,186,678	67.4 %	2,467 %	53.6 %
Europe	837,472,045	10.6 %	750,045,495	89.6 %	614 %	13.7 %
Latin America / Carib.	664,099,841	8.4 %	543,396,621	81.8 %	2,907 %	9.9 %
North America	374,226,482	4.7 %	349,572,583	93.4 %	223 %	6.4 %
Middle East	268,302,801	3.4 %	211,796,760	78.9 %	6,378 %	3.9 %
Oceania / Australia	43,602,955	0.5 %	31,191,971	71.5 %	309 %	0.6 %
WORLD TOTAL	7,934,462,631	100.0 %	5,473,055,736	69.0 %	1,416 %	100.0 %

**Internet Users in the World
by Geographic Regions - 2022**



Source: Internet World Stats - www.internetworldstats.com/stats.htm
Basis: 5,385,798,406 Internet users estimated in June 30, 2022
Copyright © 2022, Miniwatts Marketing Group

Figura. Utilizatorii de Internet în lume - împărțirea pe zone geografice

Iunie 2022 | Sursa:
<http://www.internetworldstats.com/>

Prezent

Number of internet users worldwide as of 2022, by region
(in millions)

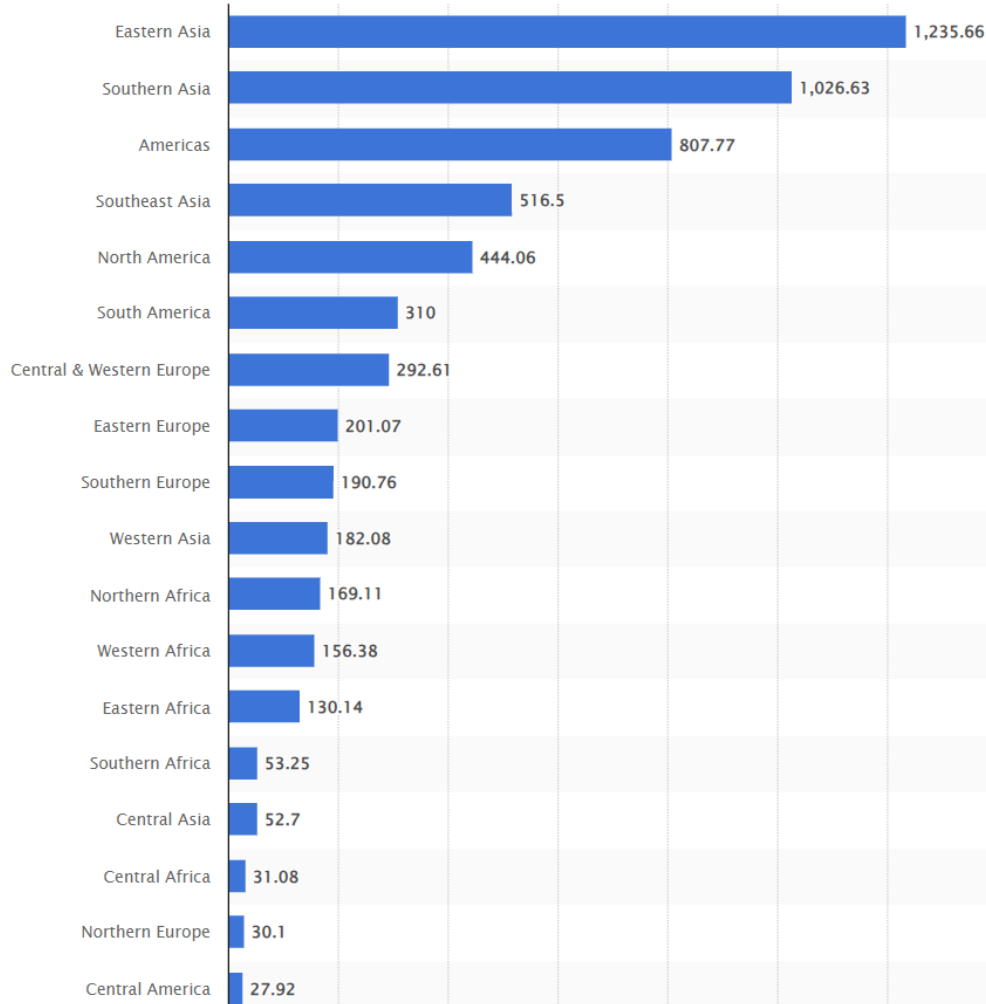


Figure. Internet users in the world
- geographically divided
June 2022| Source:
<https://www.statista.com/statistics/249562/number-of-worldwide-internet-users-by-region/>

Prezent

Number of internet users worldwide as of 2023, by subregion
(in millions)

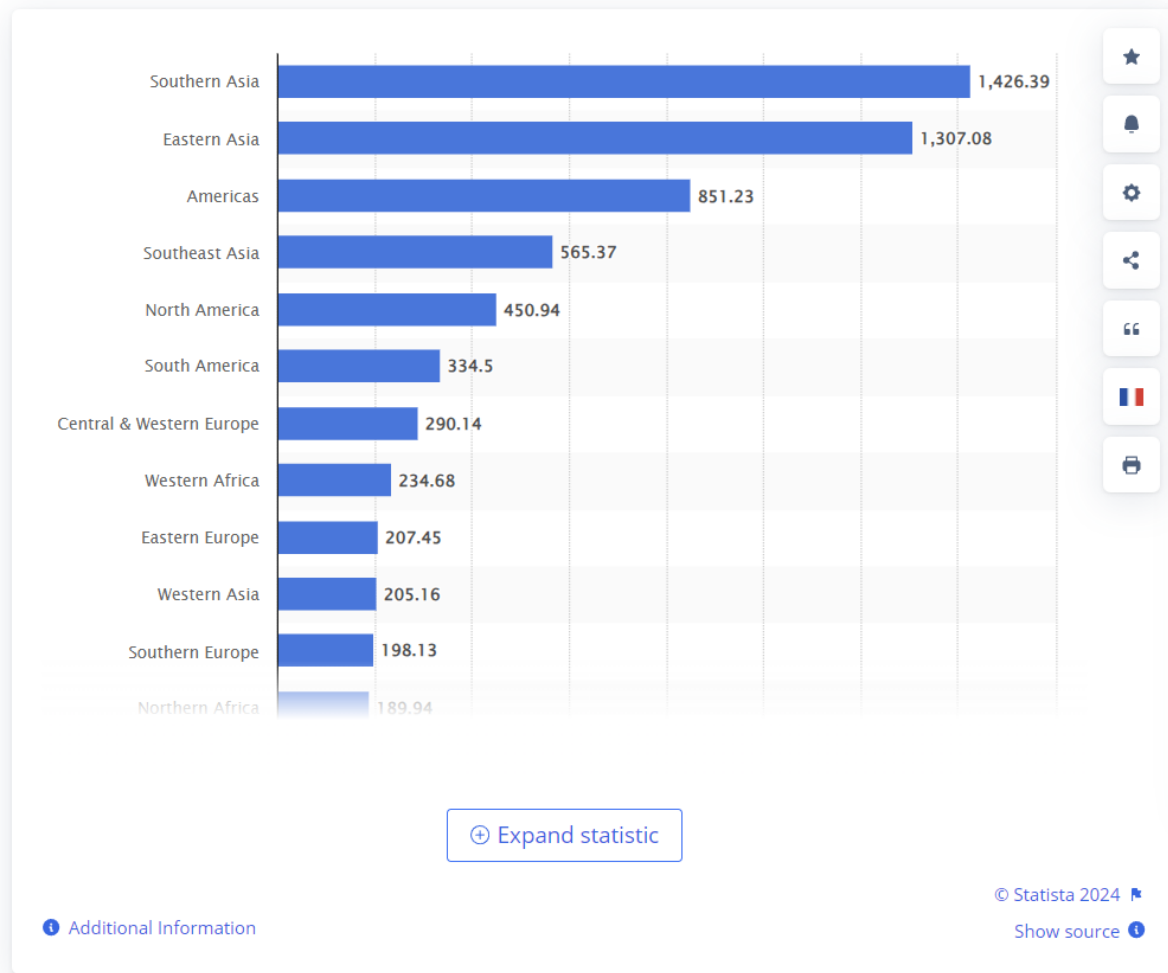
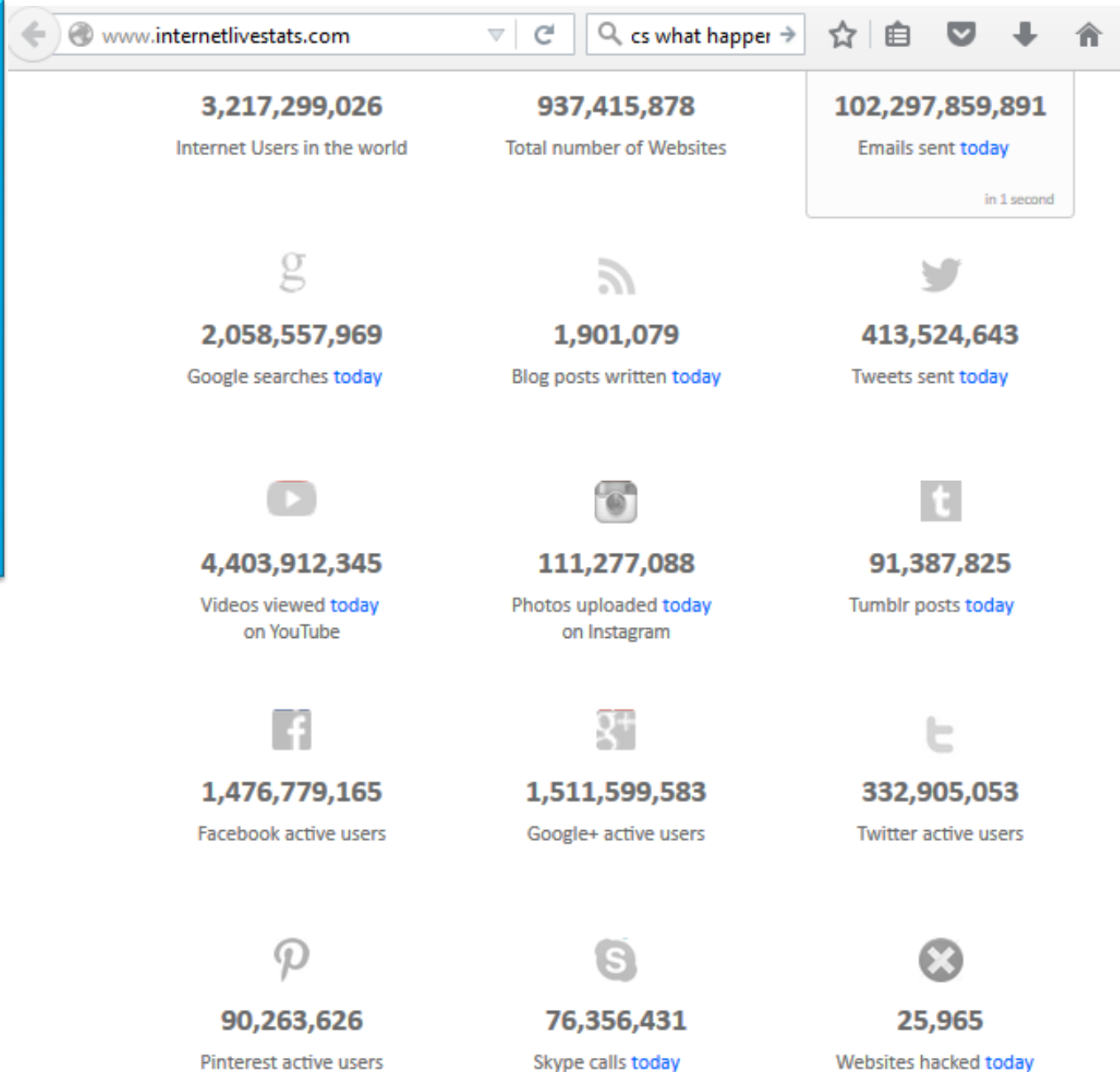
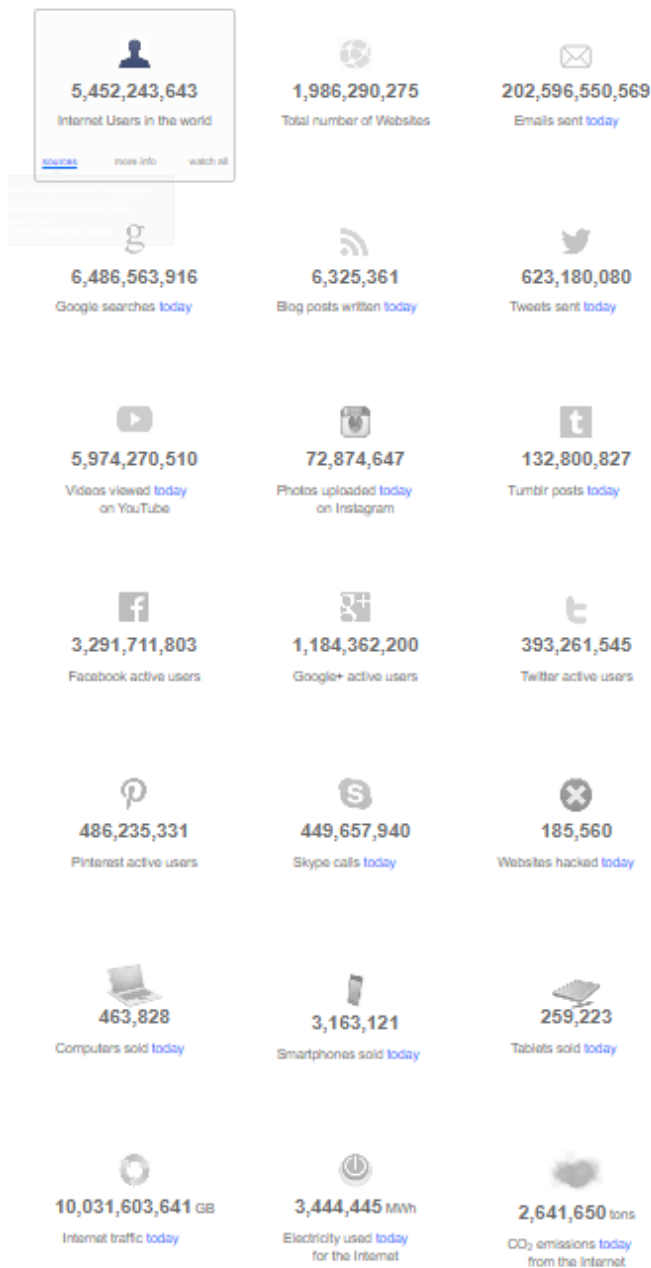


Figure. Internet users in the world
- geographically divided
June 2024| Source:
<https://www.statista.com/statistics/249562/number-of-worldwide-internet-users-by-region/>

2015



2022



Internet Live Stats has been cited by



The Official World Wide Web Anniversary Site
World Wide Web Consortium (W3C)
World Wide Web Foundation

Prezent

Cursul de **Rețele de Calculatoare** – Saptamânile 1-13-

Viitor

Cursul de **Rețele de Calculatoare** – Saptamâna 14-

Resurse

- Internet Society: <https://www.isoc.org>
- FNC – Federal Networking Council:
https://www.nitrd.gov/fnc/Internet_res.html
- InternetWorld Stats: <https://www.internetworldstats.com>
- Internet Systems Consortium: <http://www.isc.org>
- Internet Assigned Numbers Authority (IANA): <http://www.iana.org/>
- Cisco Systems: <http://www.cisco.com>
- <https://www.nethistory.info>
- <https://www.caida.org/>
- https://news.netcraft.com/archives/web_server_survey.html
- <https://www.swivel.com/>
- <https://www.facebook.com/cisconetworkingacademy> :)
- <https://www.internetlivestats.com/>
- <https://www.worldometers.info/>
- <https://www.internetsociety.org/globalinternetreport/?gclid=CN6a2Y2wqMgCFUfkwgodZ8gL9A>
- <https://www.coursera.org/learn/quantum-computing-for-everyone-an-introduction>